Companion to Bird & de Moor, Algebra of Programming

§0. Laws from Relation Algebra

§0.1. Adjunctions

(0.1a)

	F⊣G	F's type	monad FG	1⊑FG	GF⊑1	F preserves ∪	G preserves ∩	FGF=F	GFG=G
1	◁⊣▷	A→ A⊗A	◁▷=𝟙	𝟙⊑◁▷	▷◁⊑𝟙	(R ∪ S)◁= R◁ ∪ S◁	(R∩S)▷= R▷∩S▷	◁▷◁=◁	▷◁▷=▷
2	⊸⊣⊸	A→𝕀	⊸⊸=⊤	𝟙⊑⊸⊸	⊸⊸⊑𝟙	(R ∪ S)⊸= R⊸ ∪ S⊸	(R∩S)⊸= R⊸∩S⊸	⊸⊸⊸=⊸	⊸⊸⊸=⊸
3	°⊣°	(A→B)→ (B→A)	R°° = R	R°° = R	R°° = R	(R ∪ S)° = S° ∪ R°	(R ∩ S)° = R° ∩ S°	R°° = R	R°° = R
4	⊸◁⊣▷⊸	(X⊗A→Y)→ (X→Y⊗A)	𝟙	(⊸◁⊗𝟙) (𝟙⊗▷⊸)=𝟙	(𝟙⊗⊸◁) (▷⊸⊗𝟙)=𝟙	=	=	=	=
5	▷⊸⊣⊸◁	(X→Y⊗A)→ (X⊗A→Y)	𝟙	(𝟙⊗⊸◁) (▷⊸⊗𝟙)=𝟙	(⊸◁⊗𝟙) (𝟙⊗▷⊸)=𝟙	=	=	=	=
6	Δ⊣∩	(A→B)→ (A→B)²	R ∩ R = R	R ∩ R = R	R ∩ S ⊑ R	Δ(R ∪ S)= ΔR ∪ ΔS	(R∩T)∩(S∩U)= (R∩S)∩(T∩U)	R ∩ R = R	R ∩ R = R
7	∪⊣Δ	(A→B)²→ (A→B)	(R,S)↦ (R ∪ S,R ∪ S)	R ⊑ R ∪ S	R ∪ R = R	(R ∪ T) ∪ (S ∪ U)= (R ∪ S) ∪ (T ∪ U)	Δ(R∩S)= ΔR∩ΔS	R ∪ R = R	R ∪ R = R
8	𝟘⊣!	{*}→ (A→B)			𝟘 ⊑ R				
9	·S⊣/S	(A→B)→ (A→C)	S/S	R⊑(RS)/S	(T/S)S⊑T	(S ∪ T) R = (S R) ∪ (T R)	(R ∩ R') / S = (R / S) ∩ (R' / S)	((RS)/S)S =RS	((T/S)S)/S =T/S
10	S·⊣S\	(B→C)→ (A→C)	S\S	R⊑S\(SR)	S(S\T)⊑T	R (S ∪ T) = (R S) ∪ (R T)	S \ (R ∩ R') = (S \ R) ∩ (S \ R')	S(S\(SR)) =SR	S\(S(S\T)) =S\T
11	R∩⊣R⇒	(A→B)→ (A→B)	X↦R⇒(X∩R)	X⊑R⇒(X∩R)	R∩(R⇒Y)⊑Y	R∩(X ∪ Y)= (R∩X) ∪ (R∩Y)	R ⇨ S ∩ T = (R ⇨ S) ∩ (R ⇨ T)	R∩(R⇒(X∩R)) =X∩R	R⇒(R∩(R⇒Y)) =R⇒Y
12	𝒟⊣·⊤	(A→B)→ Cor A	R↦(𝒟R)⊤	R⊑(𝒟R)⊤	𝒟(A⊤)⊑A	𝒟(R ∪ S)= 𝒟R ∪ 𝒟S	(A∩B)⊤= A⊤∩B⊤	𝒟((𝒟R)⊤) =𝒟R	(𝒟(A⊤))⊤ =A⊤
13	ℛ⊣⊤·	(A→B)→ Cor B	R↦⊤(ℛR)	R⊑⊤(ℛR)	ℛ(⊤A)⊑A	ℛ(R ∪ S)= ℛR ∪ ℛS	⊤(A∩B)= ⊤A∩⊤B	ℛ(⊤(ℛR)) =ℛR	⊤(ℛ(⊤A)) =⊤A
14	·f⊣·f°	(A→B)→ (A→C)	ff°	Map(f) ⇒ 𝟙 ⊑ f f°	f°f⊑𝟙	(S ∪ T) R = (S R) ∪ (T R)	Simple(g) ⇒ (X ∩ Y) g° = (X g°) ∩ (Y g°)	ff°f=f	f°ff°=f°
15	f°·⊣f·	(A→C)→ (B→C)	ff°	Map(f) ⇒ 𝟙 ⊑ f f°	f°f⊑𝟙	R (S ∪ T) = (R S) ∪ (R T)	f(X∩Y)= fX∩fY	ff°f=f	f°ff°=f°
16	i⊣E	Map↪Rel	E	$\frac{𝟙}{∋}$:A→PA	∋:PB→B	—	—	$\frac{∋}{∋}$=𝟙	$\frac{R}{∋}$∋=R
17	·∋⊣ $\frac{}{∋}$	Map(A,PB)→ (A→B)	𝟙	$\frac{f∋}{∋}$=f	$\frac{R}{∋}$∋=R	—	—	$\frac{f∋}{∋}$∋ =f∋	$\frac{\frac{R}{∋}∋}{∋}$ =$\frac{R}{∋}$
18	$\frac{}{∋}$ ⊣·∋	(A→B)→ Map(A,PB)	𝟙	$\frac{R}{∋}$∋=R	$\frac{f∋}{∋}$=f	—	—	$\frac{\frac{R}{∋}∋}{∋}$ =$\frac{R}{∋}$	$\frac{f∋}{∋}$∋ =f∋

§0.2. Composing adjunctions

(0.2a)

		·T	T·	·g	g°·	T∩	°	⊸◁	Δ
1	·S	R / (S₁ S₂) = R / S₂ / S₁	S \ (R / W) = (S \ R) / W	R/(Sg)= (Rg°)/S	Map(f) ⇒ f (R / S) = (f R) / S	—	(Y/S)°= S°\(Y°)	(RS)^= R^(S⊗𝟙)	(R ∩ R') / S = (R / S) ∩ (R' / S)
2	S·	S \ (R / W) = (S \ R) / W	(S₁ S₂) \ R = S₂ \ (S₁ \ R)	S\(Rg°)= (S\R)g°	(g°S)\R= S\(gR)	—	(S\Y)°= Y°/S°	((S⊗𝟙)R)^= SR^	S \ (R ∩ R') = (S \ R) ∩ (S \ R')
3	·f	Map(f) ⇒ R / (f S) = (R / S) f°	T\(Rf°)= (T\R)f°	(R S)° = S° R°			(R S)° = S° R°		Simple(g) ⇒ (X ∩ Y) g° = (X g°) ∩ (Y g°)
4	f°·	Map(f) ⇒ f (R / S) = (f R) / S	(Tf°)\R= f(T\R)	—	(R S)° = S° R°		(R S)° = S° R°		f(T₁∩T₂)= fT₁∩fT₂
5	R∩	—	—	—	—	R ⇨ S ⇨ T = R ∩ S ⇨ T	(R⇒Y)°= R°⇒(Y°)	—	R ⇨ S ∩ T = (R ⇨ S) ∩ (R ⇨ T)
6	°	(Y/T)°= T°\(Y°)	(T\Y)°= Y°/T°	(R S)° = S° R°	(R S)° = S° R°	(T⇒Y)°= T°⇒(Y°)	R°° = R	(R^)°= (R°⊗𝟙) (𝟙⊗▷⊸) both bends: the ° section's display	(T₁∩T₂)°= T₁°∩T₂°
7	⊸◁	(RT)^= R^(T⊗𝟙)	((T⊗𝟙)R)^= TR^	—	—	—	(R^)°= (R°⊗𝟙) (𝟙⊗▷⊸) both bends: the ° section's display	—	—
8	∪	(S ∪ T) R = (S R) ∪ (T R)	R (S ∪ T) = (R S) ∪ (R T)	(S ∪ T) R = (S R) ∪ (T R)	R (S ∪ T) = (R S) ∪ (R T)	R ∩ (S ∪ T) = R ∩ S ∪ R ∩ T	(R ∪ S)° = S° ∪ R°		
9	0	0 R = 0	R 0 = 0	0 R = 0	R 0 = 0	R ∩ 0 = 0	0° = 0		

§0.3. Adjoint triples

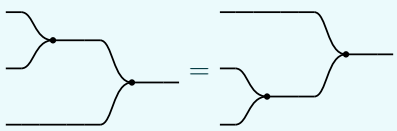
(0.3a)

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·f⊣·f°⊣/f°
 f°·⊣f·⊣f\
```

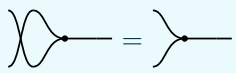
§0.4. Frobenius

Frobenius.

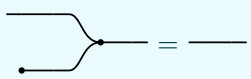
(0.4a)



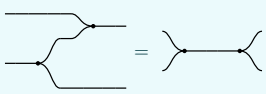
▷ associative



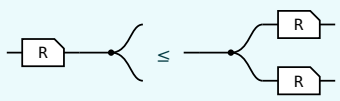
▷ commutative



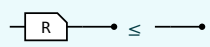
⊸ is its unit



Frobenius, one half — the other is its °

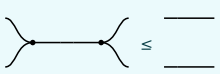


R ◁ ≤ ◁ (R ⊗ R)

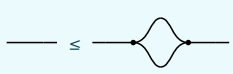


R ⊸ ≤ ⊸

(0.4b)



▷ ◁ ≤ 𝟙



𝟙 ≤ ◁ ▷

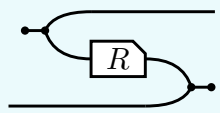
⊸ ⊸ ≤ 𝟙

𝟙 ≤ ⊸ ⊸

§0.5. ° is a contravariant 2-functor

(0.5a)

° is primitive, part of the data the first section lists. It turns both of R's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R°, not its definition:



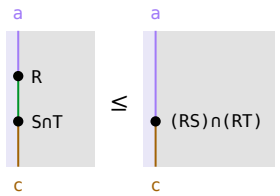
$$R^\circ=(\multimap\!\triangleleft\otimes\mathbb{1})(\mathbb{1}\otimes R\otimes\mathbb{1})(\mathbb{1}\otimes\triangleright\!\multimap)$$

where $\multimap\!\triangleleft : \mathbb{I}\to a\otimes a$ opens a pair of wires out of nothing and $\triangleright\!\multimap : b\otimes b\to\mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\multimap$ are $\triangleright^\circ$ and $\multimap^\circ$. The laws of ° are that it is a contravariant 2-functor $^\circ : \mathcal{C}^{\circ\mathrm{p}}\to\mathcal{C}$:

(i) $\mathbb{1}^\circ=\mathbb{1}$ (ii) $(RS)^\circ=S^\circ R^\circ$ (iii) $(R\otimes S)^\circ=R^\circ\otimes S^\circ$ (iv) $R\le S$ implies $R^\circ\le S^\circ$

§0.6. Meet and composition

(0.6a)

semi-distributivity, and what supplies it	picture
R (S∩T)⊑RS∩RT — the lax copy law. Equality exactly when R is single valued: the Maps section's F(R∩S)=FR∩FS.	

§0.7. Domain and range

(0.7a)

```
dom(R) ≜ 𝟙 ∩ (R R°)
```

```
Coreflexive(X) ⇒ dom(R) ⊑ X ⇔ R ⊑ X R
```

```
dom(R S) ⊑ dom(R)
```

```
dom(R ∩ S) = 𝟙 ∩ (S R°)
```

```
R entire Entire(R) ≜ dom(R) = 𝟙
```

```
R simple Simple(R) ≜ R° R ⊑ 𝟙
```

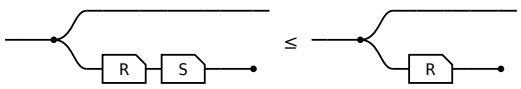
```
R a map Map(R) ≜ Entire(R) ∧ Simple(R)
```

```
Entire(R); Entire(S) ⇒ Entire(R S)
Simple(f); Simple(g) ⇒ Simple(f g)
Map(f); Map(g) ⇒ Map(f g)
```

```
Entire(R S) ⇒ Entire(R)
```

§0.8. Sliding the discard

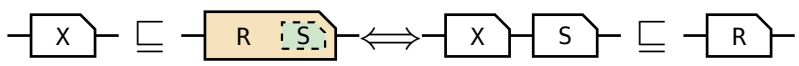
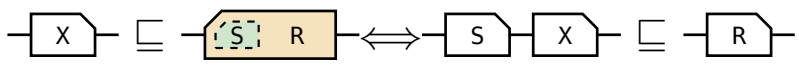
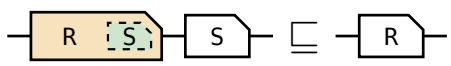
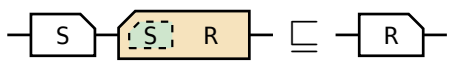
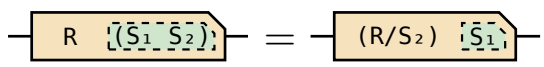
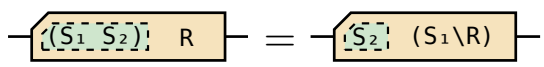
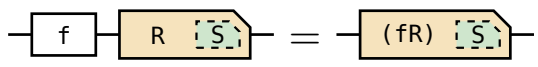
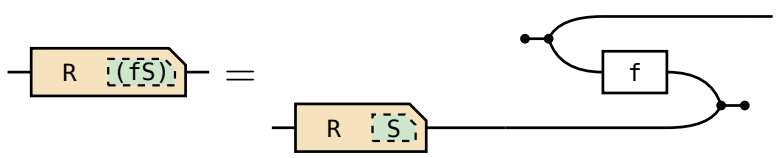
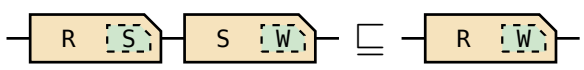
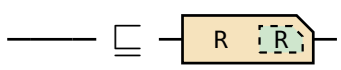
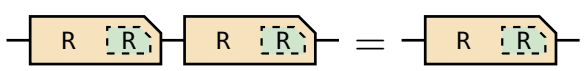
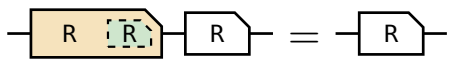
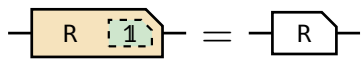
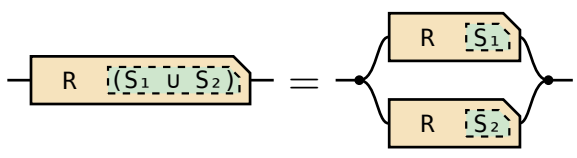
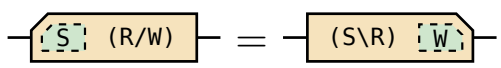
(0.8a)



S⊸⊑⊸, the lax axiom for ⊸ in the first section — the discard slides back past S

§0.9. Division

(0.9a)

1	X⊑R/S⇔XS⊑R X is any x-to-y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S, and R/S is the largest such.	
2	X⊑S\R⇔SX⊑R The mirror — divide on the left when x comes first.	
3	(R/S)S⊑R There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at S=∅: R/S is everyone, (R/S)S=∅.	
4	S (S\R)⊑R The mirror.	
5	**associate:** R/(S₁S₂)=(R/S₂)/S₁ **A friend's enemy** is two hops: divide by the far end first.	
6	(S₁S₂)\R=S₂\(S₁\R) The mirror.	
7	**maps:** f (R/S)=(fR)/S Rename x before or after dividing — the licence to write fR/S.	
8	R/(fS)=(R/S)f° Rename y: a map leaves a denominator as f° outside the box.	
9	(R/S)(S/W)⊑R/W Someone who admires all of a hate-set that already covers everyone z works for admires those people too.	
10	𝟙⊑R/R R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
11	(R/R)(R/R)=R/R R/R is the preorder **admires at least as much as**, and a preorder is idempotent. Freyd writes ⊑; with 𝟙⊑R/R above it is an equality.	
12	(R/R)R=R Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
13	R/𝟙=R Dividing by 𝟙: p's set is just {p}, so admiring all of it is admiring p.	
14	R/(S₁ ∪ S₂)=R/S₁∩R/S₂ Admiring a combined hate-set is admiring each set in full.	
15	S\(R/W)=(S\R)/W Which is why S\R/W needs no bracket.	

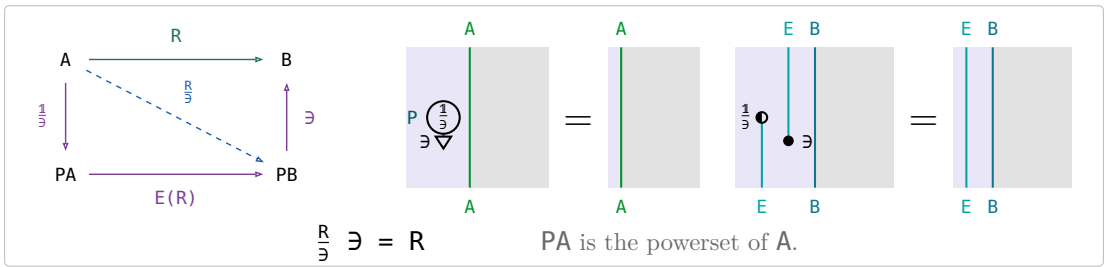
§0.10. Power allegories

(0.10a)

	law	the reading
1	$\ni_R \square = R\square$, $\ni_R = \ni_{R\,\square}$	One ∋ per object, not per arrow.
2	∋ is **thick**	**Comprehension**: every x has a set of exactly the people x admires.
3	$\frac{R}{\ni}$: A→PB, for R : A→B	convert a relation to a function. a $\frac{R}{\ni}$ ={b|a R b}
4	Map($\frac{R}{\ni}$)	
5	$\frac{R}{\ni}$ ∋ = R	reading the chosen set back through ∋ returns the relation.
6	Simple(F) ⇒ F ⊑ $\frac{F\,\ni}{\ni}$	A partial choice of sets is inside the total one.
7	the **singleton map** is monic: $\frac{\mathbb{1}}{\ni}$ ($\frac{\mathbb{1}}{\ni}$)° ⊑ 𝟙 two points with the same one-person set are the same point.	The one-person set.
8	$\frac{\ni}{\ni}$ = 𝟙	Make the set of a set, then read it back one level down.
9	**fusion:** Map(f) ⇒ $\frac{f\,R}{\ni}$ = f $\frac{R}{\ni}$	Naturality of the unit, Map(f) ⇒ f $\frac{\mathbb{1}}{\ni}$ = $\frac{\mathbb{1}}{\ni}$ E(f) . renaming a point and then taking its one-person set is taking the set and renaming inside it.
10	Map(f); 𝟙□ = ∋□ ⇒ $\frac{f}{\ni}$ = f $\frac{\mathbb{1}}{\ni}$	Rename first or take singletons first — the fusion row above at R=𝟙.
11	R□ = ∋□ ⇒ R /ₛ S = $\frac{R}{\ni}$ ($\frac{S}{\ni}$)°	x and y match when R sends x and S sends y to the same set.
12	E(R) ≜ $\frac{\ni\,R}{\ni}$	E(R): PA→PB, image of a set of A
13	$\frac{R}{\ni}$ = $\frac{\mathbb{1}}{\ni}$ E(R)	$\frac{\mathbb{1}}{\ni}$: x↦{x}
14	$\frac{S}{\ni}$ E(R) = $\frac{S\,R}{\ni}$	absorption — the monad's composition law, $\frac{S}{\ni}$ ⋄ $\frac{R}{\ni}$ = $\frac{SR}{\ni}$, §$\mathcal{A}$≅Kleisli(E) in Relation Algebra
15	E(R) = E($\frac{R}{\ni}$) union	the image of a set under R is the union of the R-sets of its points.
16	⊇ ≜ ∋ / ∋	xs⊇ys⇔∀a. ys∋a→xs∋a
17	⊆ = ⊇°	xs⊆ys⇔ys⊇xs
18	T° $\frac{T}{\ni}$ ⊑ ∈	every T-value of a point is a member of its T-set: Λ(T) is a map and Λ(T)∋=T.

§0.11. i⊣E: power transpose through singleton

(0.11a)

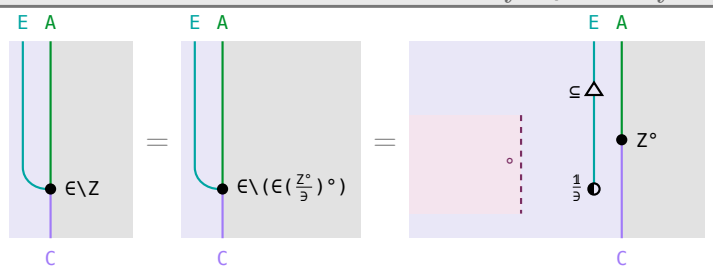


$\frac{R}{\ni}\ \ni = R$ PA is the powerset of A.

§0.12. ∈\ as a composite

$$\in \setminus Z = \subseteq (\tfrac{Z^\circ}{\ni})^\circ \qquad \textbf{(0.12a)}$$

xs is related to c by ∈\Z exactly when xs ⊆ Λ(Z°)(c), the set of every a with a Z c



1	∈ \ Z	Z=∈Λ(Z°)° — R/∋ = R
2	= ∈ \ (∈ (Z°/∋)°)	X\(Yf°)=(X\Y)f°, f a map
3	= ⊆ (Z°/∋)°	